

Diabetes Prediction from Non-Invasive Body Signals

A feature analysis and modeling roadmap for a multimodal sensor dataset

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1. Executive summary

This document analyzes a multimodal dataset of 15,720 measurement sessions collected to support diabetes detection without invasive blood tests. The dataset combines optical (near-infrared) sensor readings, wearable physiological signals (heart rate, blood volume pulse, skin conductance, wrist temperature, movement), and demographic information (age, BMI), alongside a ground-truth diabetes label derived from blood glucose measurements (diabetic if glucose exceeds 126 mg/dL).

Let me show you below of the pages, which of these non-invasive signals carry meaningful information about a person's diabetes status, so that a future predictive model can estimate diabetes risk using only sensors that do not require a blood draw or injection, for example, a wearable device or an optical skin sensor.

Key finding: features derived from a dual-wavelength near-infrared (NIR) optical sensor (660nm and 940nm) show the strongest relationship with diabetes status in this dataset, followed by heart-rate-based features from wearable sensors. Demographic features such as age and BMI, while clinically important in general populations, show negligible correlation within this specific dataset.

1.1 Headline numbers

Total measurement sessions (rows)	15,720
Total columns / features	57
Diabetic sessions	10,480 (66.7%)
Non-diabetic sessions	5,240 (33.3%)
Real-world data	8,826 rows (56.1%)
Synthetic data	6,894 rows (43.9%)
Train / validation / test split	70% / 15% / 15% (stratified)
Strongest non-invasive feature	NIR @ 940nm (corr. -0.72)

2. Background and motivation

2.1 The clinical problem

Diabetes is conventionally diagnosed and monitored using invasive blood tests: finger-prick glucose meters or laboratory blood draws that measure blood glucose concentration directly. While accurate, these methods are uncomfortable, require consumables (test strips, lancets), and are impractical for continuous or frequent monitoring.

A non-invasive alternative would estimate diabetes risk or blood glucose level using signals that can be captured externally, such as light absorption through the skin or changes in heart rate and skin conductance. If such signals correlate strongly with blood glucose or diabetes status, a wearable or handheld optical device could provide an early screening tool or continuous monitoring without needles.

2.2 Why near-infrared (NIR) optics?

Near-infrared light interacts with glucose molecules and the tissue surrounding blood vessels. When light at specific wavelengths (commonly around 660nm, red light, and 940nm, infrared) passes through or reflects off skin, the amount absorbed depends partly on the concentration of glucose in the blood and interstitial fluid. This is the same basic principle used in pulse oximetry, adapted here for glucose-related sensing.

Because the absorption signal is also influenced by skin tone, thickness, and blood perfusion, a single wavelength reading is noisy. Using the ratio between two wavelengths helps cancel out some of these confounding factors, which is why a derived 'optical ratio' feature is included in this dataset.

2.3 Why wearable physiological signals?

Diabetes affects the autonomic nervous system and cardiovascular function over time. People with diabetes can show altered resting heart rate, reduced heart rate variability, changes in blood volume pulse (BVP) waveform shape, and altered electrodermal activity (skin conductance). These signals are already captured by consumer wearables (smartwatches, fitness bands) and could provide supporting evidence alongside optical sensing.

2.4 Project goal

The goal of this analysis is to determine, using the available dataset, which non-invasive signals (optical and wearable) most strongly distinguish diabetic from non-diabetic individuals, and to propose a feature set and modeling approach for a predictive system that requires no blood sample or injection.

3. Dataset overview

3.1 Composition by source

The dataset aggregates measurement sessions from five underlying sources, combining real-world sensor recordings with statistically modeled synthetic data designed to be faithful to published correlation values.

Source dataset	Rows	Type
PhysioCGM	6,993	Real
Nature Scientific Reports – NIR Glucose	5,920	Synthetic
Kaggle Raman Diabetes	974	Synthetic
Raman Sugars	933	Real
NTNU NIR Glucose	900	Real

Table 3.1 — Row counts by underlying source dataset.

3.2 Class balance

The dataset is imbalanced toward diabetic cases (66.7% diabetic vs 33.3% non-diabetic). This should be accounted for during model training, for example using class weighting or resampling, to avoid a model that simply predicts 'diabetic' for every input.

3.3 Train / validation / test split

The dataset has been pre-partitioned into a stratified 70/15/15 split, preserving the same diabetic/non-diabetic ratio across all three partitions.

Split	Rows	Diabetic	Non-diabetic
Train (70%)	11,004	7,336 (66.7%)	7,336 (33.3%)
Validation (15%)	2,358	1,572 (66.7%)	786 (33.3%)
Test (15%)	2,358	1,572 (66.7%)	786 (33.3%)

Table 3.2 — Stratified train/validation/test partition sizes, identified by the split column.

3.4 Ground-truth label

The target label, `final_label__is_diabetic`, is derived from the blood glucose measurement: a value of 1 indicates blood glucose above 126 mg/dL (diabetic), and 0 indicates a normal reading. The diabetic group has a mean glucose of approximately 186.4 mg/dL, compared to 108.6 mg/dL for the non-diabetic group, confirming the label is correctly separating the two populations.

4. Methodology

4.1 Correlation analysis

For every numeric column, the Pearson correlation coefficient was computed against `final_label__is_diabetic` (1 = diabetic, 0 = non-diabetic). This measures the strength and direction of the linear relationship between each feature and diabetes status, on a scale from -1 to +1.

- A correlation near 0 indicates the feature does not separate diabetic from non-diabetic individuals in this dataset.
- A positive correlation means the feature value tends to be higher in diabetic individuals.
- A negative correlation means the feature value tends to be lower in diabetic individuals.
- Correlations above roughly 0.4 in magnitude are considered strong for this analysis; between 0.15 and 0.4 are considered moderate; below 0.15 are considered weak or negligible.

4.2 Group mean comparison

For each feature, the mean and standard deviation were computed separately for the diabetic group and the non-diabetic group. A large difference in means relative to the standard deviations indicates the feature is likely to be useful for separating the two classes.

4.3 Feature tiering

Based on the correlation magnitude and the practicality of measurement, features were grouped into three tiers:

- Tier 1 — Primary signal: strong correlation ($|r| > 0.4$) and derived from a single non-invasive sensing modality (dual-wavelength optical sensor).
- Tier 2 — Supporting signal: moderate correlation ($0.05 < |r| < 0.4$), derived from standard wearable sensors, useful in combination with Tier 1.
- Tier 3 — Negligible: correlation close to 0 in this dataset, despite sometimes being clinically relevant in the broader literature.

5. Tier 1 : primary optical signal

These three columns are derived from a dual-wavelength near-infrared optical sensor (660nm red light and 940nm infrared light). They represent the strongest non-invasive signals available in this dataset, with correlations between 0.49 and 0.72 in magnitude.

Dataset column	Corr.	Diabetic avg	Non-diab. avg	Interpretation
feature__nir_signal_intensity_at_940nm...	-0.72	546.5	630.2	Strongest single predictor. Infrared absorption is markedly lower in diabetic readings.
feature__nir_signal_intensity_at_660nm...	-0.67	650.1	712.5	Red-light absorption also drops in diabetic readings, consistent with the 940nm signal.
feature__optical_ratio_660nm_divided_by_940nm...	+0.49	1.20	1.13	Ratio of the two wavelengths; cancels skin tone and thickness effects. Most robust single feature for a real device.

Table 5.1 — Tier 1 features with Pearson correlation and group means.

Practical implementation: a single dual-wavelength optical sensor placed on the fingertip, earlobe, or wrist — similar in form factor to a pulse oximeter — can produce all three of these features from one reading. The standard deviation of the 940nm signal is approximately 36–42 units within each group, meaning the roughly 84-unit gap between group means (546.5 vs 630.2) represents a meaningfully large, measurable separation.

6. Tier 2 — supporting wearable signals

These columns come from standard wearable sensors: photoplethysmography (PPG) for heart rate and blood volume pulse, and electrodermal activity (EDA) sensors for skin conductance. Individually weaker than Tier 1, they can improve overall model accuracy when combined with the optical features.

Dataset column	Corr.	Diabetic avg	Non-diab. avg	Interpretation
heart_rate_avg_bpm	+0.20	80.6	74.4	Average resting heart rate is about 6 bpm higher in diabetic individuals.
heart_rate_max_bpm	+0.19	84.7	78.0	Peak heart rate shows the same upward shift as the average.
heart_rate_min_bpm	+0.20	76.6	71.0	Minimum heart rate during the session is also elevated in diabetics.
bvp_variability	-0.10	95.8	106.5	Blood volume pulse variability is reduced in diabetic readings, consistent with altered vascular tone.
bvp_max / bvp_min	± 0.09 – 0.10	see note	see note	Both the maximum and minimum of the BVP waveform show a smaller magnitude swing in diabetics.
movement_intensity_peak	-0.07	103.8	108.9	Slightly lower peak movement intensity, a mild link to activity level.
skin_conductance_avg_uS	+0.05	2.20	1.94	Average electrodermal activity is marginally higher in diabetic readings.
heart_rate_variability_bpm	+0.07	2.52	2.21	Heart rate variability is slightly higher in diabetics in this dataset.

Table 6.1 — Tier 2 features with Pearson correlation and group means. BVP max/min values are reported in raw sensor units; see note below.

Note on BVP max/min: the diabetic group shows bvp_max averaging 445.0 versus 511.7 for non-diabetic, and bvp_min averaging -500.8 versus -567.3. In both cases the diabetic group's waveform has a smaller amplitude swing (closer to zero), consistent with the reduced bvp_variability noted above.

Practical implementation: all Tier 2 features are obtainable from a consumer smartwatch or fitness band equipped with a PPG sensor (heart rate, BVP) and, less commonly, an EDA sensor (skin conductance). These add no additional hardware burden beyond what many existing wearables already include.

7. Tier 3 — features with negligible signal in this dataset

The following features are frequently cited as diabetes risk factors in the medical literature, but show correlations close to zero against the diabetic label within this specific dataset. This is an important caveat for the supervisor's awareness: it does not mean these factors are clinically unimportant in general, but that this dataset's construction (particularly its synthetic portions) does not encode a relationship between these variables and diabetes status.

Dataset column	Corr.	Diabetic avg	Non-diab. avg	Interpretation
demographic__body_mass_index...	+0.03	29.26	29.07	Clinically, BMI > 30 is a strong predictor of type 2 diabetes. This dataset shows almost no difference between groups.
demographic__volunteer_age_years...	-0.01	45.21	45.51	Clinically, risk increases sharply after age 45. No age gap is present between groups here.
physiological...SpO2...	-0.02	97.91	97.94	Blood oxygen saturation is essentially identical between groups.
physiological...skin_surface_temperature...	+0.05	33.01	32.93	Skin temperature is nearly identical between groups, though it affects calibration of the NIR sensors.
environmental...ambient_room_temperature...	+0.01	24.19	24.16	Ambient temperature during measurement shows no relationship with diabetes status, as expected.
wrist_temp_avg_C / movement_intensity_avg	≈0.01–0.02	—	—	Average wrist temperature and average movement intensity show negligible separation between groups.

Table 7.1 — Tier 3 features: negligible correlation with diabetic label in this dataset.

Recommendation: these features can still be collected during data acquisition (they are inexpensive to measure and may help contextualize sensor readings, e.g. skin temperature for NIR calibration), but they should not be treated as primary predictive inputs for a model trained on this dataset. If future data collection includes a wider age and BMI range with a clearer relationship to diabetes status, these features should be re-evaluated.

8. Recommended feature set for a non-invasive model

Based on the analysis above, the following feature set is recommended as the input to an initial non-invasive diabetes classification model. It combines the strongest available signal (Tier 1, optical) with supporting wearable signals (Tier 2), all obtainable without any blood draw or injection.

8.1 Primary inputs (required)

- feature__nir_signal_intensity_at_660nm__red_wavelength__higher_means_less_absorption
- feature__nir_signal_intensity_at_940nm__infrared_wavelength__higher_means_less_absorption
- feature__optical_ratio_660nm_divided_by_940nm__cancels_skin_thickness_and_melanin_noise

8.2 Supporting inputs (recommended)

- heart_rate_avg_bpm
- heart_rate_max_bpm
- bvp_variability
- skin_conductance_avg_uS

8.3 Target variable

- final_label__is_diabetic (binary: 1 = diabetic, 0 = non-diabetic)

This is equivalent to label__is_diabetic__1_equals_glucose_above_126_mgdL__0_equals_normal, which encodes the same clinical threshold.

9. Proposed modeling plan

9.1 Baseline models

As a starting point, two simple, interpretable models are recommended:

- Logistic regression on the recommended feature set — provides an interpretable baseline and per-feature coefficients.
- Random forest or gradient-boosted trees — captures non-linear relationships and feature interactions, and provides feature importance rankings to validate the correlation-based tiering in this report.

9.2 Handling class imbalance

Since the dataset is approximately 67% diabetic and 33% non-diabetic, class weighting (e.g. `class_weight='balanced'` in scikit-learn) or resampling techniques should be applied during training to avoid a model biased toward predicting the majority class.

9.3 Evaluation metrics

Given the class imbalance, accuracy alone is insufficient. The following metrics are recommended on the held-out test set:

- Sensitivity (recall for diabetic class) — the ability to correctly identify diabetic individuals, critical for a screening tool.
- Specificity (recall for non-diabetic class) — the ability to avoid false alarms.
- ROC-AUC and precision-recall AUC — overall discriminative ability independent of classification threshold.
- Confusion matrix — to inspect the balance of false positives vs false negatives.

9.4 Validation strategy

The provided stratified 60/20/20 train/validation/test split should be used as-is: the training set for model fitting, the validation set for hyperparameter tuning, and the test set held out until final evaluation only.

9.5 Next steps

- Train and evaluate the baseline logistic regression and random forest models on the recommended Tier 1 + Tier 2 feature set.
- Compare model-derived feature importances against the correlation-based tiering in this report to confirm consistency.
- Assess whether adding Tier 3 (demographic) features changes performance, despite their low individual correlations — interaction effects are possible.
- If results are promising, evaluate the model on real-world-only data (excluding synthetic rows) to estimate performance on genuine sensor readings.

10. Limitations and caveats

- Approximately 44% of the dataset is synthetic, generated to be statistically consistent with published correlations. Findings should be validated on real-world data before drawing clinical conclusions.
- The dataset's label is derived from a single blood glucose threshold (126 mg/dL) and does not capture other diagnostic criteria (e.g. HbA1c, oral glucose tolerance test).
- Correlation does not imply causation; the relationships described here indicate statistical association within this dataset, not a validated physiological mechanism for this specific population.
- Demographic features (age, BMI) showing negligible correlation here contradicts general clinical literature, likely reflecting limitations in how the synthetic data was generated rather than a true absence of relationship in real populations.
- This analysis is exploratory and intended to guide feature selection for a modeling phase; it does not constitute a validated diagnostic tool.

11. Conclusion

This analysis demonstrates that a dual-wavelength near-infrared optical sensor, combined with basic wearable vitals (heart rate and blood volume pulse variability), provides the strongest available non-invasive signal for distinguishing diabetic from non-diabetic individuals in this dataset. The proposed feature set and modeling plan offer a concrete path toward a prototype non-invasive diabetes screening model, with appropriate caveats regarding the synthetic portion of the data and the need for real-world validation.